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EC466/966: Estimation and Inference in Econometrics

Answer key-Exercise # 2

1. Please check EC966_466hw2sol.m or EC466_966hw2sol.mlx for the answer.

2. Suppose that you have the following regression model

$$\mathbf{y} = \mathbf{X}\beta + \epsilon,$$

where $\mathbf{y}$ is the $n \times 1$ vector of the dependent variable, $\mathbf{X}$ is the $n \times K$ vector of regressors, β is the $K \times 1$ vector of the model parameters, and ϵ is the $n \times 1$ vector of the error term. Assume that Classical Linear Regression Model assumptions hold. We estimate the model by the OLS estimator, and obtain $\mathbf{b}$ that is the estimate for β.

(a) Show that $\hat{\mathbf{y}} = \mathbf{P}\mathbf{y}$ where $\mathbf{P} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-\mathbf{1}}\mathbf{X}'$.

We use the definitions of $\hat{\mathbf{y}} = \mathbf{X}\mathbf{b}$ and $\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-\mathbf{1}}\mathbf{X}'\mathbf{y}$; therefore we write that

$$\hat{\mathbf{y}} = \mathbf{X}\mathbf{b} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-\mathbf{1}}\mathbf{X}'\mathbf{y} = \mathbf{P}\mathbf{y}.$$

(b) Show that $\mathbf{e} = \mathbf{M}\mathbf{y} = \mathbf{M}\epsilon$ where $\mathbf{M} = \mathbf{I_n} - \mathbf{P}$ and $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$.

First, we write that

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{P}\mathbf{y} = (\mathbf{I_n} - \mathbf{P})\mathbf{y} = \mathbf{M}\mathbf{y}.$$

Next, we use the identity for $\mathbf{y} = \mathbf{X}\beta + \epsilon$ and write that

$$\mathbf{e} = \mathbf{M}\mathbf{y} = \mathbf{M}(\mathbf{X}\beta + \epsilon) = \mathbf{M}\mathbf{X}\beta + \mathbf{M}\epsilon.$$

Note that $\mathbf{M}\mathbf{X} = (\mathbf{I_n} - \mathbf{P})\mathbf{X} = \mathbf{X} - \mathbf{P}\mathbf{X} = \mathbf{X} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-\mathbf{1}}\mathbf{X}'\mathbf{X} = \mathbf{0}$. So, we conclude that

$$\mathbf{e} = \mathbf{M}\mathbf{X}\beta + \mathbf{M}\epsilon = \mathbf{M}\epsilon.$$

(c) Show that SSR($\mathbf{b}$) $= \epsilon'\mathbf{M}\epsilon$.

We know that SSR($\mathbf{b}$) $= \mathbf{e}'\mathbf{e}$. By using the identity in the above Question 2(b) such that $\mathbf{e} = \mathbf{M}\epsilon$.

$$SSR(\mathbf{b}) = (\mathbf{M}\epsilon)'\mathbf{M}\epsilon = \epsilon'\mathbf{M}'\mathbf{M}\epsilon.$$

Note that $\mathbf{M}$ is a symmetric matrix, since $\mathbf{M}' = (\mathbf{I_n} - \mathbf{P})' = \mathbf{I_n'} - \mathbf{P}' = \mathbf{I_n'} - (\mathbf{X}(\mathbf{X}'\mathbf{X})^{-\mathbf{1}}\mathbf{X}')' = \mathbf{I_n} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-\mathbf{1}}\mathbf{X}' = \mathbf{M}$. Also, $\mathbf{M}$ is an idempotent matrix since

$$\mathbf{MM} = (\mathbf{I_n} - \mathbf{P})(\mathbf{I_n} - \mathbf{P}) = \mathbf{I_n} - 2\mathbf{P} + \mathbf{PP} \quad = \mathbf{I_n} - 2\mathbf{P} + \mathbf{P} = \mathbf{I_n} - \mathbf{P},$$

where $\mathbf{PP} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-\mathbf{1}}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-\mathbf{1}}\mathbf{X}' = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-\mathbf{1}}\mathbf{X}' = \mathbf{P}$. By using $\mathbf{M}$ being a symmetric and idempotent matrix, we can write that

$$SSR(\mathbf{b}) = \epsilon'\mathbf{M}'\mathbf{M}\epsilon = \epsilon'\mathbf{MM}\epsilon = \epsilon'\mathbf{M}\epsilon.$$

(d) Show that $s^2 = \mathbf{e}'\mathbf{e}/(n-K)$ is an unbiased estimator for $\sigma^2 = E(\epsilon_i^2|\mathbf{X})$.

To show that s^2 is an unbiased estimator for σ^2, we need to prove that $E(s^2|\mathbf{X}) = \sigma^2$.

$$\begin{aligned}
E(s^2|\mathbf{X}) &= (n-K)^{-1}E(\mathbf{e}'\mathbf{e}|\mathbf{X}) = (n-K)^{-1}E(\epsilon'\mathbf{M}\epsilon|\mathbf{X}) \\
&= (n-K)^{-1}E(tr(\epsilon'\mathbf{M}\epsilon)|\mathbf{X}) = (n-K)^{-1}E(tr(\mathbf{M}\epsilon\epsilon')|\mathbf{X}) \\
&= (n-K)^{-1}tr(E(\mathbf{M}\epsilon\epsilon'|\mathbf{X})) = (n-K)^{-1}tr(\mathbf{M}E(\epsilon\epsilon'|\mathbf{X})) \\
&= (n-K)^{-1}tr(\mathbf{M}\sigma^2\mathbf{I_n}) = \sigma^2(n-K)^{-1}tr(\mathbf{M}) \\
&= \sigma^2(n-K)^{-1}tr(\mathbf{I_n} - \mathbf{P}) = \sigma^2(n-K)^{-1}(n - tr(\mathbf{P})) \\
&= \sigma^2(n-K)^{-1}(n - tr(\mathbf{X}(\mathbf{X}'\mathbf{X})^{-\mathbf{1}}\mathbf{X}')) \\
&= \sigma^2(n-K)^{-1}(n - tr((\mathbf{X}'\mathbf{X})^{-\mathbf{1}}\mathbf{X}'\mathbf{X})) \\
&= \sigma^2(n-K)^{-1}(n - tr(\mathbf{I_K})) \\
&= \sigma^2.
\end{aligned}$$